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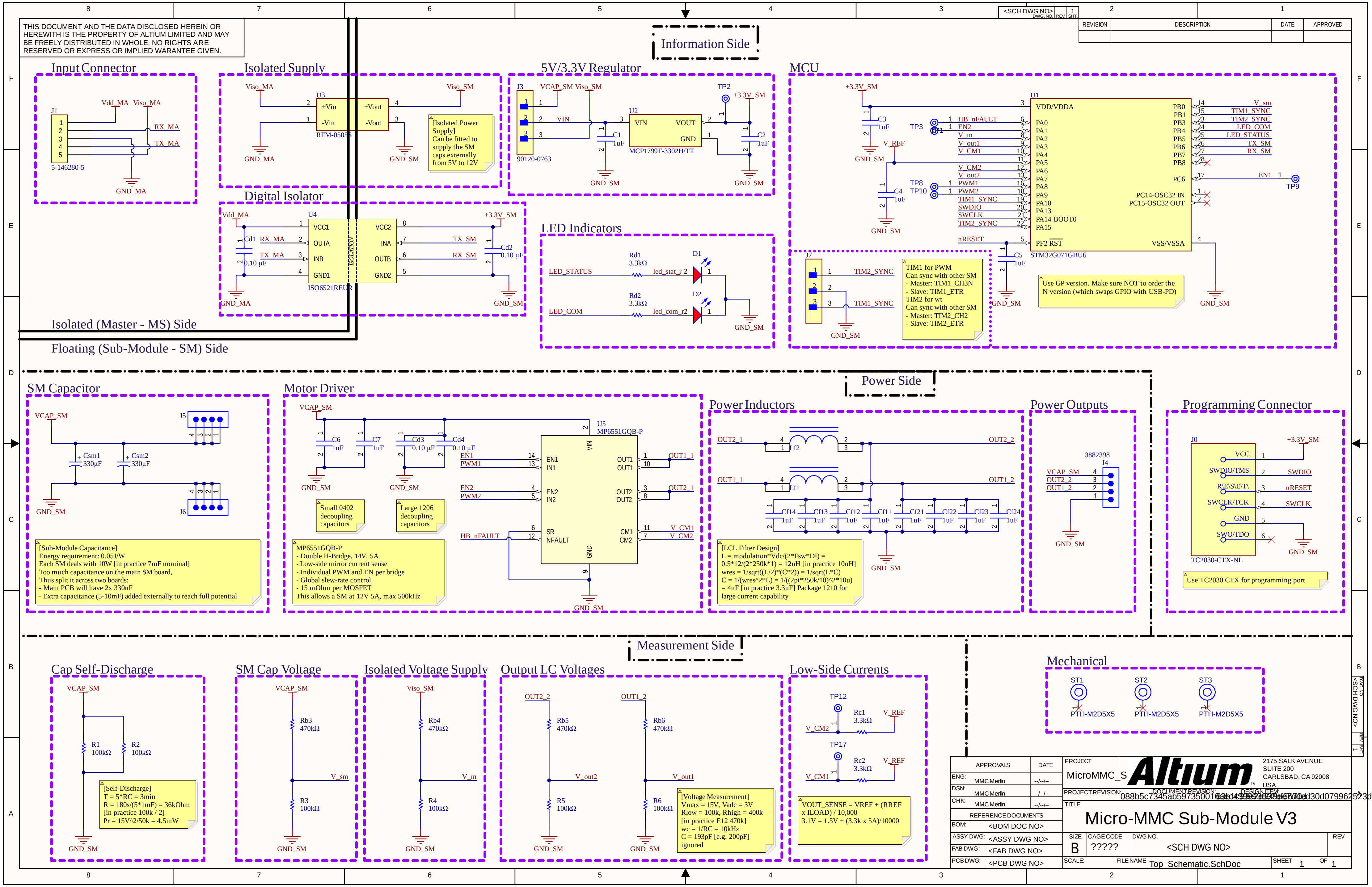

Information Side
Input Connector
J1
Vdd_MA Viso_MA
RX_MA
TX_MA
5-146280-5
GND_MA
Isolated Supply
Viso_MA
U3
+Vin
-Vin
+Vout
-Vout
RFM-0505S
Viso_SM
GND_MA
GND_SM
5V/3.3V Regulator
J3
VCAP_SM Viso_SM
VIN
90120-0763
U2
VIN
VOUT
GND
MCP1799T-3302H/TT
C1 1uF
TP2
+3.3V_SM
C2 1uF
GND_SM
MCU
+3.3V_SM
C3 1uF
GND_SM
TP3
TP1
V_REF
HB_nFAULT
EN2
V_m
V_out1
V_CM1
V_CM2
V_out2
TP8
TP10
PWM1
PWM2
TIM1_SYNC
SWDIO
SWCLK
TIM2_SYNC
nRESET
C4 1uF
C5 1uF
U1
VDD/VDDA
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA13
PA14-BOOT0
PA15
PF2 RST
VSS/VSSA
PB0
PB1
PB3
PB4
PB5
PB6
PB7
PB8
PC6
PC14-OSC32 IN
PC15-OSC32 OUT
STM32G071GBU6
V_sm
TIM1_SYNC
TIM2_SYNC
LED_COM
LED_STATUS
TX_SM
RX_SM
EN1
TP9
GND_SM
J7
TIM2_SYNC
TIM1_SYNC
GND_SM
Digital Isolator
Vdd_MA
U4
VCC1
OUTA
INB
GND1
VCC2
INA
OUTB
GND2
ISOLATION
ISO6521REUR
Cd1 0.10 µF
RX_MA
TX_MA
+3.3V_SM
TX_SM
RX_SM
Cd2 0.10 µF
GND_MA
GND_SM
LED Indicators
LED_STATUS
Rd1 3.3kΩ
led_stat_r
D1
LED_COM
Rd2 3.3kΩ
led_com_r
D2
GND_SM
Isolated (Master - MS) Side
Floating (Sub-Module - SM) Side
Power Side
SM Capacitor
VCAP_SM
J5
Csm1 330µF
Csm2 330µF
GND_SM
J6
Motor Driver
VCAP_SM
C6 1uF
C7 1uF
Cd3 0.10 µF
Cd4 0.10 µF
GND_SM
EN1
PWM1
EN2
PWM2
HB_nFAULT
U5
MP6551GQB-P
VIN
EN1
IN1
EN2
IN2
SR
NFAULT
GND
OUT1
OUT1
OUT2
OUT2
CM1
CM2
OUT1_1
OUT2_1
V_CM1
V_CM2
GND_SM
Power Inductors
OUT2_1
Lf2
OUT2_2
OUT1_1
Lf1
OUT1_2
Cf14 1uF
Cf13 1uF
Cf12 1uF
Cf11 1uF
Cf21 1uF
Cf22 1uF
Cf23 1uF
Cf24 1uF
GND_SM
Power Outputs
3882398
J4
VCAP_SM
OUT2_2
OUT1_2
GND_SM
Programming Connector
J0
+3.3V_SM
VCC
SWDIO/TMS
R\E\S\E\T\
SWCLK/TCK
GND
SWO/TDO
SWDIO
nRESET
SWCLK
TC2030-CTX-NL
GND_SM
Measurement Side
Cap Self-Discharge
VCAP_SM
R1 100kΩ
R2 100kΩ
GND_SM
SM Cap Voltage
VCAP_SM
Rb3 470kΩ
V_sm
R3 100kΩ
GND_SM
Isolated Voltage Supply
Viso_SM
Rb4 470kΩ
V_m
R4 100kΩ
GND_SM
Output LC Voltages
OUT2_2
Rb5 470kΩ
V_out2
R5 100kΩ
OUT1_2
Rb6 470kΩ
V_out1
R6 100kΩ
GND_SM
Low-Side Currents
TP12
V_CM2
Rc1 3.3kΩ
V_REF
TP17
V_CM1
Rc2 3.3kΩ
V_REF
Mechanical
ST1
ST2
ST3
PTH-M2D5X5
PTH-M2D5X5
PTH-M2D5X5

[Isolated Power Supply]
Can be fitted to supply the SM caps externally from 5V to 12V

Use GP version. Make sure NOT to order the N version (which swaps GPIO with USB-PD)

TIM1 for PWM
Can sync with other SM
- Master: TIM1_CH3N
- Slave: TIM1_ETR
TIM2 for wt
Can sync with other SM
- Master: TIM2_CH2
- Slave: TIM2_ETR

Small 0402 decoupling capacitors

Large 1206 decoupling capacitors

[Sub-Module Capacitance]
Energy requirement: 0.05J/W
Each SM deals with 10W [in practice 7mF nominal]
Too much capacitance on the main SM board,
Thus split it across two boards:
- Main PCB will have 2x 330uF
- Extra capacitance (5-10mF) added externally to reach full potential

MP6551GQB-P
- Double H-Bridge, 14V, 5A
- Low-side mirror current sense
- Individual PWM and EN per bridge
- Global slew-rate control
- 15 mOhm per MOSFET
This allows a SM at 12V 5A, max 500kHz

[LCL Filter Design]
L = modulation*Vdc/(2*Fsw*DI) = 0.5*12/(2*250k*1) = 12uH [in practice 10uH]
wres = 1/sqrt((L/2)*(C*2)) = 1/sqrt(L*C)
C = 1/(wres^2*L) = 1/(2pi*250k/10)^2*10u) = 4uF [in practice 3.3uF] Package 1210 for large current capability

Use TC2030 CTX for programming port

[Self-Discharge]
T = 5*RC = 3min
R = 180s/(5*1mF) = 36kOhm
[in practice 100k / 2]
Pr = 15V^2/50k = 4.5mW

[Voltage Measurement]
Vmax = 15V, Vadc = 3V
Rlow = 100k, Rhigh = 400k
[in practice E12 470k]
wc = 1/RC = 10kHz
C = 193pF [e.g. 200pF]
ignored

VOUT_SENSE = VREF + (RREF x ILOAD) / 10,000
3.1V = 1.5V + (3.3k x 5A)/10000

REVISION	DESCRIPTION	DATE	APPROVED

<SCH DWG NO> DWG. NO. REV. 1 SHT.

APPROVALS		DATE
ENG:	MMC Merlin	--/--/--
DSN:	MMC Merlin	--/--/--
CHK:	MMC Merlin	--/--/--

REFERENCE DOCUMENTS
BOM: <BOM DOC NO>
ASSY DWG: <ASSY DWG NO>
FAB DWG: <FAB DWG NO>
PCB DWG: <PCB DWG NO>

PROJECT
MicroMMC_SM
PROJECT REVISION: DOCUMENT REVISION: DESIGN ITEM: [illegible]

2175 SALK AVENUE
SUITE 200
CARLSBAD, CA 92008
USA

TITLE
Micro-MMC Sub-Module V3

SIZE: B
CAGE CODE: ?????
DWG NO.: <SCH DWG NO>
REV
SCALE:
FILE NAME: Top_Schematic.SchDoc
SHEET 1 OF 1